

Computability

Eric Shen

July 2025

1 Introduction

The idea behind this handout is that oftentimes it may be more efficient to solve geometry problems by evaluating how *computable* given sub-problems are. In geometry problems one often finds conditions which they want to prove or conjecture to be true, but they may not yet know how to prove. A useful perspective on such problems is determining whether there is a way to make these conditions more amenable to computation. For example, the coaxiality lemma translates coaxiality conditions into a power of a point computation, while spiral similarity theory relates many ratio conditions to concyclicities. Though these lemmas are very useful on their own (and will be discussed throughout this handout), the emphasis of this handout is not on learning these individual lemmas but on making you more fluent in translating geometric conditions into computable terms.

Some things we will talk about today:

1. Some conditions are easy to compute. When a problem only involves such conditions, you can shift your focus from doing the dirty computations, towards understanding how to set up the computation.
2. There are a lot of more advanced lemmas that help you compute other conditions. We'll cover some of them here, and showcase how they help you understand when you've reduced your problem to something computable.

The mantra for today is:

Reduce your problem into computable steps!

2 Easily Computable Conditions

When given a lot of conditions which are easy to compute, you can evaluate whether a proposition is true based on whether or not it 1) seems true in various different cases and 2) is easily verifiable to be true based on your setup. Thus for such conditions, you can focus your job on making your setup more computationally amenable.

A condition that lends itself readily to computation is **angle chasing**. Here's an example. Given a triangle ABC , the incenter I , circumcenter O , and orthocenter H all work nicely with angles; we have that $\angle XYZ$ is a rational linear combination of $\{\angle A, \angle B, \angle C, \pi\}$ when two of X, Y, Z are in $\{A, B, C\}$ and the other is in $\{I, O, H\}$. Thus any equalities between them is a matter of checking that the coefficients of a linear equation match in $\mathbb{R}/2\pi\mathbb{R}$, up to the linear dependence $\angle A + \angle B + \angle C = \pi$.

Example 2.1. (Fact 5) Given a triangle ABC , let I be its incenter and let $AI \cap (ABC) = M$. Then Fact 5 states M is the circumcenter of I , and to do this it suffices to check $\angle BIM = \angle IBM$. But $\angle BIM = \pi - \angle AIB$ and $\angle IBM = \angle IBC + \angle CBM = \angle IBC + \angle CAI$. Now all these angles are easily computable, so we know this equality has to turn out to be true after a computation.

Another condition that lends itself readily to computation is *linearity*. Angle chasing was already an example of translating angle conditions to linear equations. But linear equations show up more generally: for example, if a collection of points satisfy elementary vector equations (e.g. $\vec{H} + 2\vec{O} = 3\vec{G}$), then checking another vector equation is a matter of solving a system of equations. Area conditions, or distances from a point to a given line, are other conditions which are “linear” in the point (i.e. they are functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which satisfy that for $k \in \mathbb{R}$, $P, Q \in \mathbb{R}^2$, one has $f(kP + (1 - k)Q) = kf(P) + (1 - k)f(Q)$).

Example 2.2. (EGMO 2025) Let ABC be an acute triangle. Points B, D, E , and C lie on a line in this order and satisfy $BD = DE = EC$. Let M and N be the midpoints of AD and AE , respectively. Suppose triangle ADE is acute, and let H be its orthocentre. Points P and Q lie on lines BM and CN , respectively, such that D, H, M , and P are concyclic and pairwise different, and E, H, N , and Q are concyclic and pairwise different. Prove that P, Q, N , and M are concyclic.

1. Let $PM \cap QN = R$, and suppose (PMH) and (QNH) intersect at H and another point S . Prove that it suffices to show H, R, S collinear.
2. Let $HS \cap (MSN) = T$. Prove $TM = TN$.
3. Prove T is actually the circumcenter of AMN .
4. We’re now effectively done! Let O be the circumcenter of ADE and L the midpoint of DE ; use homotheties to relate $\vec{A}, \vec{R}, \vec{L}$ and $\vec{A}, \vec{T}, \vec{O}$ together. Now finish by manipulating the appropriate linear equation.

This example showcases how computability heuristics helps solve problems: it doesn’t immediately solve any big steps, but it does make many smaller steps a lot more digestible.

A **big** warning: today’s class is just a soft skill! Just because this class gives you heuristics for when you should believe something is true, that doesn’t mean it is true unless you actually prove it. So you need to prove all the menial computations or phrase your argument in rigorous terms, or else it will not be complete (and you’ll lose your points).

3 Geometry Translate™

Harder problems ($>$ IMO 1) will usually require more than just elementary angle chases and coordinate bashes. Here are some more advanced techniques for translating geometric conditions into computational ones. Of course this list isn’t complete, but it lists some of my favourite methods.

3.1 Perpendicularity Lemma

Here’s a simple but very useful lemma.

Lemma 3.1. Consider points A, B, C, D in the plane. Then $AC \perp BD$ iff $AB^2 + CD^2 = AD^2 + BC^2$.

Proof. The Pythagorean theorem. □

Here’s a powerful showcase of the Perpendicularity lemma.

Example 3.2. Consider a hexagon $ABCDEF$. Then the perpendiculars from A, C, E to BD, DF, FB (respectively) concur iff the perpendiculars from B, D, F to CE, EA, AC (respectively) concur.

Proof. Verify both conditions are equivalent to $AB^2 + CD^2 + EF^2 = BC^2 + DE^2 + FA^2$. $\square$

Here's another example.

Example 3.3. (MOP HW 2025 G2) Let ABC be a triangle, and let $D \in BC, E \in CA, F \in AB$ such that $ABDE, ACDF$ are cyclic. Suppose (AEF) intersects BC at two points X and Y . Then prove $DX = DY$.

1. Let O be the circumcenter of AEF . Prove it suffices to show $DO \perp BC$.
2. Use power of a point and the perpendicularity lemma to finish.

3.2 Cross Ratios and the Prism Lemma

Recall that for 4 points $z_1, z_2, z_3, z_4 \in \mathbb{C}$, their *cross-ratio* is defined as

$$[z_1, z_2; z_3, z_4] := \frac{z_1 - z_3}{z_1 - z_4} \div \frac{z_2 - z_3}{z_2 - z_4}.$$

The cross ratio being real is equivalent to z_1, z_2, z_3, z_4 being concyclic/collinear, which is usually the case for the cross ratios we will care about. We call a 4-tuple (z_1, z_2, z_3, z_4) *harmonic* when their cross ratio is -1 .

Note that cross ratios satisfy that

$$[z_1, z_2; z_3, z_4] = [z_2, z_1; z_3, z_4]^{-1} = [z_2, z_1; z_4, z_3] = [z_3, z_4; z_1, z_2].$$

Cross ratios can also be defined for lines:

Definition 3.4. Consider five distinct points A, B, C, D, P , where A, B, C, D are collinear. Then define the cross ratio $[PA, PB; PC, PD]$ as

$$[PA, PB; PC, PD] := [A, B; C, D] = \frac{\sin(\angle APC)}{\sin(\angle APD)} \div \frac{\sin(\angle BPC)}{\sin(\angle BPD)}.$$

Note that the second equality implies that the cross ratio $[PA, PB; PC, PD]$ is independent of the choice of A, B, C, D , and only depends on the lines PA, PB, PC, PD themselves. Using this, one can show:

Lemma 3.5. Fix two lines l, l' . Consider four other lines l_1, l_2, l_3, l_4 , all passing through a point P . Denote $A_i := l \cap l_i$, $A'_i := l' \cap l_i$. Then

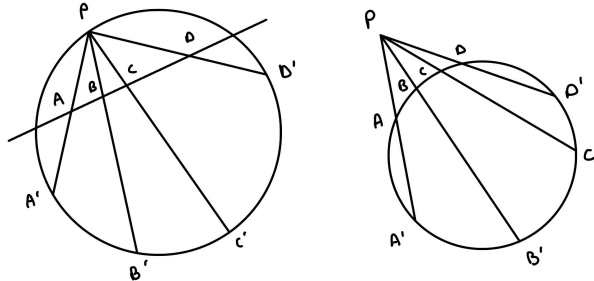
$$[A_1, A_2; A_3, A_4] = [A'_1, A'_2; A'_3, A'_4].$$

One can also compute (using the complex number definition):

Lemma 3.6. Cross-ratios are preserved under inversion. That is, consider points P, A, B, C, D , and let A', B', C', D' be the images of A, B, C, D under inversion about P by arbitrary radius. Then

$$[A, B; C, D] = [A', B'; C', D'].$$

Exercise 3.7. Prove that $[A, B; C, D] = [A', B'; C', D']$ in the examples below.



Cross-ratios can be used for a lot of useful computations!

Example 3.8. Consider a triangle ABC , and points $D, G \in BC$, $E \in CA$, and $F \in AB$, such that AD, BE, CF concur at P and E, F, G are collinear. Let $AD \cap EF = T$. Then

$$[G, D; B, C] =^A [G, T; F, E] =^P [G, D; C, B] = [G, D; B, C]^{-1}.$$

So $[G, D; B, C] = \pm 1$; since it's clearly not 1, it equals -1 . This implies that

$$\frac{GB}{GC} = \frac{DB}{DC}.$$

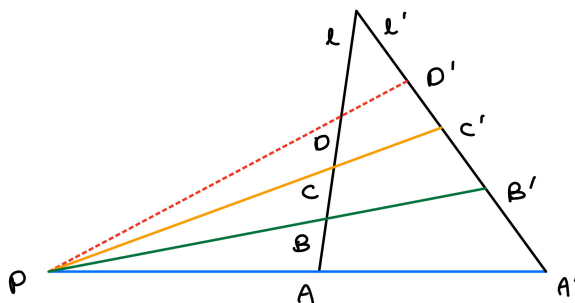
One useful property about cross-ratios is that if $[z_1, z_2; z_3, z_4] = [z'_1, z'_2; z'_3, z'_4]$, then $z_1 = z'_1$. Thus you can prove that two points are the same by comparing some cross-ratios. For example, this idea leads to the useful *prism lemma*:

Lemma 3.9. (*Prism Lemma*) Consider two lines l, l' and points $A, B, C, D \in l, A', B', C', D' \in l'$, such that AA', BB', CC' concur at a point P . Then DD' also passes through P iff

$$[A, B; C, D] = [A', B'; C', D'].$$

Proof. Suppose PD intersects l' at D'' . Then $[A', B'; C', D''] = [A, B; C, D]$, so $D' = D''$ iff $[A', B'; C', D'] = [A, B; C, D]$. $\square$

The Prism lemma allows us to switch between collinearity statements and cross-ratio equalities.



Example 3.10. Consider a triangle ABC and let points D, E be on segment BC . Let I_1, I_2, I_3, I_4 be the incenters of ABD, ABE, ADC, AEC , respectively. Prove that $I_1 I_4, I_2 I_3, BC$ concur.

1. Show it suffices to show that $[B, I_1; I_2, I] = [I, I_3; I_4, C]$.
2. Prove this equality using a *rotation*.

3.3 Spiral Similarities

We've already discussed earlier in camp how spiral similarities can work really well with lengths. Here are two highlights:

1. Consider a triangle ABC and a point $P \in (ABC)$. Then for $D \in AB$, $E \in AC$, one has that $ADEP$ is cyclic iff

$$\frac{DB}{EC} = \frac{PB}{PC}.$$

2. Fix two distinct non-parallel lines l, l' and let $A = l \cap l'$. Consider varying points $B \in l, C \in l'$ such that the lengths AB and AC satisfy some fixed linear relation (i.e. $x \cdot AB + y \cdot AC = z$). Then all such (ABC) pass through a fixed point.

For more details on why these facts are true (and how to use them), see the Spiral Sims Canada IMO Camp 2025 handout. We'll also see some examples later today.

3.4 Relating Points, and Circumcenter Spam

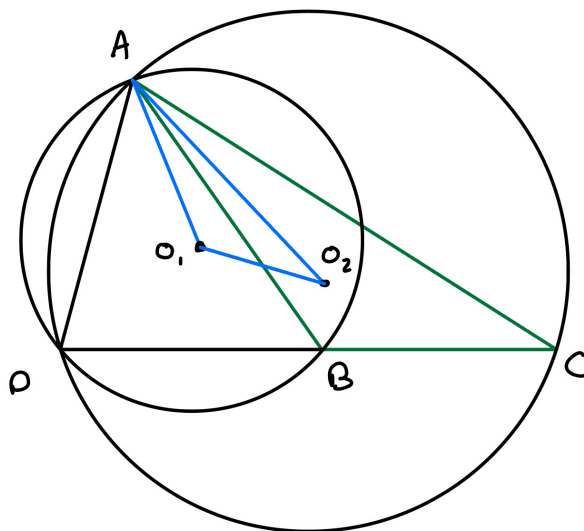
Let's now take a step back from these computational techniques, and discuss a method that helps you manage computations better. Oftentimes what makes geometric objects difficult to compute is that they seem to be "floating"; for example a point might be defined by some weird condition and thus be difficult to pin down. There are a variety of techniques and ideas to deal with such issues. Today, however, we'll focus on one: points defined as an intersection of two circles. For such points, it might be useful to consider:

1. the radical axis between the two circles (so we can put the point on a line), or
2. the circumcenters of the circles.

We'll focus on that second idea. Adding circumcenters is especially useful when one of the intersections between the circles is easy to pin down, but the other isn't; we then can at least understand it as the reflection across the lines passing through the circumcenters! One lemma that is especially useful in this scenario is Salmon's theorem.

Theorem 3.11. (Salmon) Let ABC be a triangle, and let $D \in BC$. Let O_1 and O_2 be the circumcenters of ADB and ADC , respectively. Then $AO_1O_2 \sim ABC$.

Proof. Observe $AO_1B \sim AO_2C$ (as both are isosceles with apex angle $2\angle ADB$), which implies the result. $\square$



Here's a nice example, which you all may remember.

Example 3.12. (2024 G4) Let $ABCD$ be a convex quadrilateral with AB parallel to CD and $AB < CD$. Lines AD and BC intersect at a point P . Point $X \neq C$ on the circumcircle of triangle ABC is such that $PC = PX$. Point $Y \neq D$ on the circumcircle of triangle ABD is such that $PD = PY$. Lines AX and BY intersect at Q .

Assume that $Q \neq P$. Prove that PQ is parallel to AB .

Here's a solution, which follows Christopher Li's. Let $\omega_1, \omega_2, \omega_3, \omega_4$ be the circumcircles of ABC , ABD , PCX , PDY respectively, O_i the center of ω_i for $i \in \{1, 2, 3, 4\}$, and let l be the line through P parallel to AB (and CD).

1. Suppose $l \cap AX = Q_X$. Prove that $Q_X \in \omega_3$.
2. Prove that it suffices to show $O_3O_4 \perp AB$.
3. Prove that it suffices to show that

$$\frac{O_3P}{O_3O_1} = \frac{O_4P}{O_4O_2}.$$

4. Show this equality using Salmon or otherwise.

We can see clearly in this question how the circumcenters help tie together Q_X and Q_Y , which are otherwise "floating".

3.5 More Techniques

There are a plethora of other hybrid techniques to learn which can help turn geometric conditions computational. Here are some others which come to mind:

1. The Ratio lemma. Mira94 has a great handout on it here.
2. The "forgotten Coaxiality lemma". I don't know any standard sources for this; here is one I found online.
3. Linearity of Power of a Point. Mira94 discusses this (and more) in the handout here.

However, don't forget: the point of this handout is not just to teach some computational techniques, but to really home in the mindset that it is useful to approach geometry problems with the mindset of making conditions computable. This not only lets you do more computations, but helps you see how to reduce the problem in general.

4 A Hard Example

Example 4.1. (MOP HW 2025 G5) Let ADE be a triangle with circumcenter O . P and Q are on DE such that $AP = PD$ and $AQ = QE$. Let $(APQ) \cap (ADE) = \{A, K\}$. Moreover, consider $B \in AE$, $C \in AD$ such that

$$\angle AKB = \angle AKC = 90^\circ - \angle DAE.$$

Prove that B, O, C collinear.

A short disclaimer: there are some configuration issues with B and C , so let's suppose K is on the arc DE of (ADE) not containing A , and B and C are on the respective line segments.

Here are two paths, one without inversion, one with inversion. Both exemplify the themes of today's lecture in different ways!

First is the path without starting with inversion.

1. Let H be the orthocenter of ADE , and let H_D, H_E be the reflections of H across AE, AD . Prove that B is on KH_D .
2. Prove that H is on BC , and hence it suffices to prove that K is the intersection of the reflections of OH across the three sides of ADE .
3. Let X, Y be the feet of the altitudes from K to AD, AE respectively. Prove that it suffices to show that OH and XY are parallel.

These last two steps were really just invoking Simson & Steiner line theory. It's useful stuff; remember it!

4. Now, let $(ODE) \cap AD, AE = Q', P'$ respectively. Show that $AP'Q'K$ is cyclic.

Here's now the strategy.

5. Consider two arbitrary points U, V in the same plane as A, D, E . Suppose the perpendiculars from U, V to AE intersect AD at U_1, V_1 (respectively), while the perpendiculars from U, V to AD intersect AE at U_2, V_2 (respectively). Prove that the *orientation* of UV depends only on the (signed) ratio of $|U_1V_1|$ to $|U_2V_2|$.
6. By applying this lemma and doing some spiral similarity computations, finish.

Here's the path that starts with $(\sqrt{bc})$ inversion. In this case, the problem reads as follows.

Consider a triangle APQ with incenter O . Let $D = OQ \cap (APQ)$, $E = OP \cap (APQ)$, and $K = DE \cap PQ$. Let the perpendicular from K to AE intersect AD at B , and let the perpendicular from K to AD intersect AE at C . Prove $(ABCO)$ is cyclic.

1. Prove this is indeed the problem you get after $\sqrt{bc}$ inversion about A in ADE .

Now, we'll make the cyclic condition into a length computation as follows. Consider $Y \in AB, Z \in AC$ such that $KA = KY = KZ$. Then $AOYZ$ is cyclic with center K .

2. Prove that it suffices to show $BY/CZ = OY/OZ$.
3. Show $OYZ \sim FED$.
4. Consider $B' \in AB, C' \in AC$ such that $KB = KB', KC = KC'$. Moreover, let $PD \cap QE = T$. Prove it suffices to show $AB'KC' \sim FDTE$.

These last few steps have all just been length manipulations in an attempt to demystify the length condition.

5. Prove that it suffices to show that KA and FT intersect on (DEF) .
6. Prove the above using a cross-ratio computation.

I want to really emphasize the problem-solving mantra being illuminated by these two examples. In both solutions we use a step (the orientation lemma in the first, the spiral similarity in the second), where a hybrid technique is specifically used to change a more unwieldy condition into one more tractable. *This* is really what you should watch out for: understanding that these hybrid tools can be used to “data simplifiers”, in translating your conditions into more computable ones.

5 Problems

The selection of practice problems here is pretty arbitrary: I've mostly chosen problems in which it helps to have a "computability" mindset, but honestly for a proficient user this mindset helps for a good majority of geometry problems. So perhaps the best practice for this skill is to take geometry problems you would do ordinarily, and approach them with a more computational, reductive bent.

Below, problems are organized into the categories of Part A (IMO 1-2 level), Part B (IMO 2+ level), and Part C (IMO 3+ level). $\diamond$ denotes problems which I think fit the handout's theme well, while $\star$ problems are problems which I think are particularly nice.

5.1 Part A

IMO 1-2 level problems.

1. $\diamond$ In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.
2. Let ABC be a fixed acute triangle inscribed in a circle ω with center O . A variable point X is chosen on minor arc AB of ω , and segments CX and AB meet at D . Denote by O_1 and O_2 the circumcenters of triangles ADX and BDX , respectively. Determine all points X for which the area of triangle OO_1O_2 is minimized.
3. $\diamond$ Let ABC be a triangle and let P be a point in its interior. Denote by D, E, F the feet of the perpendiculars from P to the lines BC, CA, AB , respectively. Suppose that

$$AP^2 + PD^2 = BP^2 + PE^2 = CP^2 + PF^2.$$

Denote by I_A, I_B, I_C the excenters of the triangle ABC . Prove that P is the circumcenter of the triangle $I_AI_BI_C$.

4. Let acute scalene triangle ABC have orthocenter H and altitude AD with D on side BC . Let M be the midpoint of side BC , and let D' be the reflection of D over M . Let P be a point on line $D'H$ such that lines AP and BC are parallel, and let the circumcircles of $\triangle AHP$ and $\triangle BHC$ meet again at $G \neq H$. Prove that $\angle MHG = 90^\circ$.
5. Let ABC be an acute triangle with orthocenter H . Points A_1, B_1, C_1 are chosen in the interiors of sides BC, CA, AB , respectively, such that $\triangle A_1B_1C_1$ has orthocenter H . Define $A_2 = \overline{AH} \cap \overline{B_1C_1}$, $B_2 = \overline{BH} \cap \overline{C_1A_1}$, and $C_2 = \overline{CH} \cap \overline{A_1B_1}$.
Prove that triangle $A_2B_2C_2$ has orthocenter H .
6. $\star$ Let ABC be a triangle for which there exists an interior point F such that $\angle AFB = \angle BFC = \angle CFA$. Let the lines BF and CF meet the sides AC and AB at D and E respectively. Prove that

$$AB + AC \geq 4DE.$$

7. Let ABC be an acute triangle with orthocenter H . Let G be the point such that the quadrilateral $ABGH$ is a parallelogram. Let I be the point on the line GH such that AC bisects HI . Suppose that the line AC intersects the circumcircle of the triangle GCI at C and J . Prove that $IJ = AH$.
8. $\diamond$ Let ABC be an acute-angled triangle and let D, E , and F be the feet of altitudes from A, B , and C to sides BC, CA , and AB , respectively. Denote by ω_B and ω_C the incircles of triangles BDF and CDE , and let these circles be tangent to segments DF and DE at M and N , respectively. Let line MN meet circles ω_B and ω_C again at $P \neq M$ and $Q \neq N$, respectively. Prove that $MP = NQ$.
9. Let ABC be a triangle with $AB \neq AC$ and circumcenter O . The bisector of $\angle BAC$ intersects BC at D . Let E be the reflection of D with respect to the midpoint of BC . The lines through D and E perpendicular to BC intersect the lines AO and AD at X and Y respectively. Prove that the quadrilateral $BXCY$ is cyclic.

5.2 Part B

IMO 2+ level problems.

1. Let ABC be a triangle with orthocenter H and let P be the second intersection of the circumcircle of triangle AHC with the internal bisector of the angle $\angle BAC$. Let X be the circumcenter of triangle APB and Y the orthocenter of triangle APC . Prove that the length of segment XY is equal to the circumradius of triangle ABC .
2. Let ABC be a triangle. Find all points P on segment BC satisfying the following property: If X and Y are the intersections of line PA with the common external tangent lines of the circumcircles of triangles PAB and PAC , then

$$\left(\frac{PA}{XY}\right)^2 + \frac{PB \cdot PC}{AB \cdot AC} = 1.$$

3. $\diamond$ Let O and H be the circumcenter and orthocenter, respectively, of an acute scalene triangle ABC . The perpendicular bisector of $\overline{AH}$ intersects $\overline{AB}$ and $\overline{AC}$ at X_A and Y_A respectively. Let K_A denote the intersection of the circumcircles of triangles $OX_A Y_A$ and BOC other than O .

Define K_B and K_C analogously by repeating this construction two more times. Prove that K_A , K_B , K_C , and O are concyclic.

4. Let $ABCC_1B_1A_1$ be a convex hexagon such that $AB = BC$, and suppose that the line segments AA_1 , BB_1 , and CC_1 have the same perpendicular bisector. Let the diagonals AC_1 and A_1C meet at D , and denote by ω the circle ABC . Let ω intersect the circle A_1BC_1 again at $E \neq B$. Prove that the lines BB_1 and DE intersect on ω .
5. Let BC be a fixed segment in the plane, and let A be a variable point in the plane not on the line BC . Distinct points X and Y are chosen on the rays $CA^\rightarrow$ and $BA^\rightarrow$, respectively, such that $\angle CBX = \angle YCB = \angle BAC$. Assume that the tangents to the circumcircle of ABC at B and C meet line XY at P and Q , respectively, such that the points X , P , Y and Q are pairwise distinct and lie on the same side of BC . Let Ω_1 be the circle through X and P centred on BC . Similarly, let Ω_2 be the circle through Y and Q centred on BC . Prove that Ω_1 and Ω_2 intersect at two fixed points as A varies.
6. $\star$ Let $ABCDE$ be a convex pentagon with $CD = DE$ and $\angle EDC \neq 2 \cdot \angle ADB$. Suppose that a point P is located in the interior of the pentagon such that $AP = AE$ and $BP = BC$. Prove that P lies on the diagonal CE if and only if $[BCD] + [ADE] = [ABD] + [ABP]$, where $[XYZ]$ denotes the area of a given triangle XYZ .
7. Let $ABCD$ be a cyclic quadrilateral whose sides have pairwise different lengths. Let O be the circumcenter of $ABCD$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet $\overline{AC}$ at B_1 and D_1 , respectively. Let O_B be the center of the circle which passes through B and is tangent to $\overline{AC}$ at D_1 . Similarly, let O_D be the center of the circle which passes through D and is tangent to $\overline{AC}$ at B_1 . Assume that $\overline{BD_1} \parallel \overline{DB_1}$. Prove that O lies on the line $\overline{O_B O_D}$.

5.3 Part C

IMO 3+ level problems.

1. Let P be a point on the circumcircle of acute triangle ABC . Let D, E, F be the reflections of P in the A -midline, B -midline, and C -midline. Let ω be the circumcircle of the triangle formed by the perpendicular bisectors of AD, BE, CF .
Show that the circumcircles of $\triangle ADP, \triangle BEP, \triangle CFP$, and ω share a common point.

2. Convex quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$. Point H is the foot of the perpendicular from A to BD . Points S and T lie on sides AB and AD , respectively, such that H lies inside triangle SCT and

$$\angle CHS - \angle CSB = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ.$$

Prove that line BD is tangent to the circumcircle of triangle TSH .

3. Let the sides AD and BC of the quadrilateral $ABCD$ (such that AB is not parallel to CD) intersect at point P . Points O_1 and O_2 are circumcenters and points H_1 and H_2 are orthocenters of triangles ABP and CDP , respectively. Denote the midpoints of segments O_1H_1 and O_2H_2 by E_1 and E_2 , respectively. Prove that the perpendicular from E_1 on CD , the perpendicular from E_2 on AB and the lines H_1H_2 are concurrent.
4. Let $ABCDEF$ be a convex hexagon all of whose sides are tangent to a circle ω with centre O . Suppose that the circumcircle of triangle ACE is concentric with ω . Let J be the foot of the perpendicular from B to CD . Suppose that the perpendicular from B to DF intersects the line EO at a point K . Let L be the foot of the perpendicular from K to DE . Prove that $DJ = DL$.
5. Let ABC be a scalene triangle with circumcircle Γ . Let M be the midpoint of BC . A variable point P is selected in the line segment AM . The circumcircles of triangles BPM and CPM intersect Γ again at points D and E , respectively. The lines DP and EP intersect (a second time) the circumcircles to triangles CPM and BPM at X and Y , respectively. Prove that as P varies, the circumcircle of $\triangle AXY$ passes through a fixed point T distinct from A .
6. ★ An equilateral pentagon $AMNPQ$ is inscribed in triangle ABC such that $M \in \overline{AB}$, $Q \in \overline{AC}$, and $N, P \in \overline{BC}$. Let S be the intersection of $\overrightarrow{MN}$ and $\overrightarrow{PQ}$. Denote by ℓ the angle bisector of $\angle MSQ$. Prove that $\overline{OI}$ is parallel to ℓ , where O is the circumcenter of triangle ABC , and I is the incenter of triangle ABC .
7. ◇ ★ Let ABC be a triangle with incenter I . Points K and L are chosen on segment BC such that the incircles of $\triangle ABK$ and $\triangle ABL$ are tangent at P , and the incircles of $\triangle ACK$ and $\triangle ACL$ are tangent at Q . Prove that $IP = IQ$.
8. Let D be an interior point of the acute triangle ABC with $AB > AC$ so that $\angle DAB = \angle CAD$. The point E on the segment AC satisfies $\angle ADE = \angle BCD$, the point F on the segment AB satisfies $\angle FDA = \angle DBC$, and the point X on the line AC satisfies $CX = BX$. Let O_1 and O_2 be the circumcenters of the triangles ADC and EXD , respectively. Prove that the lines BC, EF , and O_1O_2 are concurrent.
9. Let ABC be a triangle with incenter I , and whose incircle is tangent to $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ at D , E , F , respectively. Let K be the foot of the altitude from D to $\overline{EF}$. Suppose that the circumcircle of $\triangle AIB$ meets the incircle at two distinct points C_1 and C_2 , while the circumcircle of $\triangle AIC$ meets the incircle at two distinct points B_1 and B_2 . Prove that the radical axis of the circumcircles of $\triangle BB_1B_2$ and $\triangle CC_1C_2$ passes through the midpoint M of $\overline{DK}$.
10. Points A, V_1, V_2, B, U_2, U_1 lie fixed on a circle Γ , in that order, and such that $BU_2 > AU_1 > BV_2 > AV_1$.
Let X be a variable point on the arc V_1V_2 of Γ not containing A or B . Line XA meets line U_1V_1 at C , while line XB meets line U_2V_2 at D . Let O and ρ denote the circumcenter and circumradius of $\triangle XCD$, respectively.
Prove there exists a fixed point K and a real number c , independent of X , for which $OK^2 - \rho^2 = c$ always holds regardless of the choice of X .
11. Let ABC be an acute-angled scalene triangle with incircle ω and circumcircle Γ . Suppose ω touches line BC at D and the tangent to Γ at A meets line BC at T . Two circles passing through A and T tangent to ω meet line AD again at X and Y .
Prove that $BXCY$ is a trapezium.

6 Solutions to Selected Examples

(EGMO 2025) Let ABC be an acute triangle. Points B, D, E , and C lie on a line in this order and satisfy $BD = DE = EC$. Let M and N be the midpoints of AD and AE , respectively. Suppose triangle ADE is acute, and let H be its orthocentre. Points P and Q lie on lines BM and CN , respectively, such that D, H, M , and P are concyclic and pairwise different, and E, H, N , and Q are concyclic and pairwise different. Prove that P, Q, N , and M are concyclic.

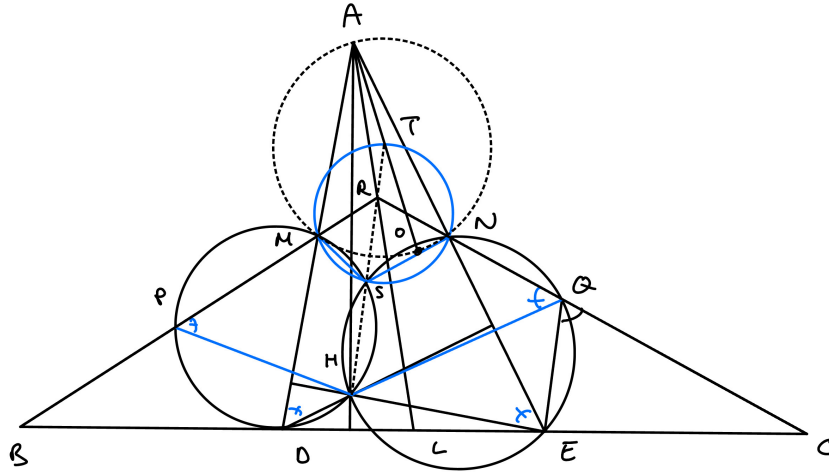
Solution 6.1. Let $PM \cap QN = R$, $(PMH) \cap (QNH) = S$, and $HS \cap (MSN) = T$. Also, let O be the circumcenter of ADE and L the midpoint of DE . By radical axis it suffices to show H, R, T collinear. Now

$$\angle TMN = \angle TSN = \pi - \angle HSN = \angle NEH = \angle OED = \angle ODE.$$

Similarly $\angle TNM = \angle OED$, so T is the circumcenter of AMN . Thus $\vec{T} = \frac{1}{2}(\vec{A} + \vec{O})$. Now by the homothety taking RMN to RBC , we find $5\vec{R} = 3\vec{A} + 2\vec{L}$. Now, let G be the centroid of ADE . Then note

$$\begin{aligned} 5\vec{R} &= 3\vec{A} + 2\vec{L} \\ 3\vec{G} &= \vec{A} + 2\vec{L} \\ &= \vec{H} + 2\vec{O} \\ 5\vec{R} &= 2\vec{A} + \vec{H} + 2\vec{O} \\ &= 4\vec{T} + \vec{H}. \end{aligned}$$

Thus H, R, T are collinear, as desired.

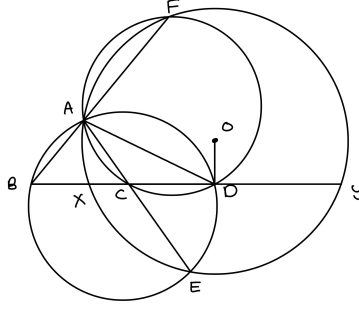


(MOP HW 2025 G2) Let ABC be a triangle, and let $D \in BC, E \in CA, F \in AB$ such that $ABDE, ACDF$ are cyclic. Suppose (AEF) intersects BC at two points X and Y . Then prove $DX = DY$.

Solution 6.2. Let O be the circumcenter of AEF ; then it suffices to show $OD \perp BC$ (as $OX = OY$). But

$$\begin{aligned} BO^2 - CO^2 &= (BO^2 - AO^2) - (CO^2 - AO^2) \\ &= BF \cdot BA - CE \cdot CA \\ &= BD \cdot BC - CD \cdot BC \\ &= BD^2 - CD^2 \end{aligned}$$

so we are done by the Perpendicularity lemma.



Consider a triangle ABC and let points D, E be on segment BC . Let I_1, I_2, I_3, I_4 be the incenters of ABD, ABE, ADC, AEC , respectively. Prove that I_1I_4, I_2I_3, BC concur.

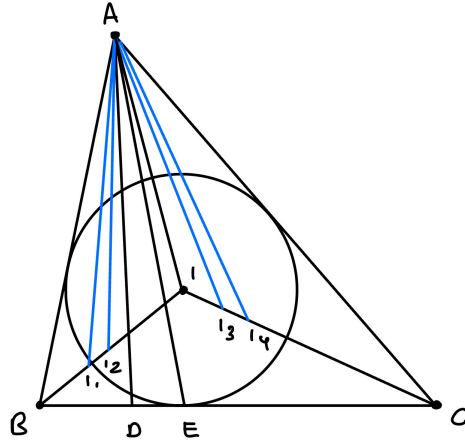
Solution 6.3. Let I be the incenter of ABC . By the Prism lemma, it suffices to show $[B, I_1; I_2, I] = [C, I_4; I_3, I]$. Now observe

$$\angle BAI = \angle I_1AI_3 = \angle I_2AI_4 = \angle IAC = \frac{1}{2}\angle A.$$

Thus by rotating about A by an angle of $\angle A/2$, we get

$$[B, I_1; I_2, I] = [AB, AI_1; AI_2, AI] = [AI, AI_3; AI_4, AC] = [I, I_3; I_4, C].$$

Now it is clear $[I, I_3; I_4, C] = [C, I_4; I_3, I]$, so we are done.



(2024 G4) Let $ABCD$ be a convex quadrilateral with AB parallel to CD and $AB < CD$. Lines AD and BC intersect at a point P . Point $X \neq C$ on the circumcircle of triangle ABC is such that $PC = PX$. Point $Y \neq D$ on the circumcircle of triangle ABD is such that $PD = PY$. Lines AX and BY intersect at Q .

Assume that $Q \neq P$. Prove that PQ is parallel to AB .

Here's a solution, which follows Christopher Li's. Let $\omega_1, \omega_2, \omega_3, \omega_4$ be the circumcircles of ABC, ABD, PCX, PDY respectively, O_i the center of ω_i for $i \in \{1, 2, 3, 4\}$, and let l be the line through P parallel to AB (and CD). Moreover, let $Q_X = AX \cap l$ and $Q_Y = BY \cap l$. Note that $\angle PQ_XX = \angle Q_XAB = \angle BCX$, so $Q_X \in \omega_3$. Similarly $Q_Y \in \omega_4$. Since we wish to prove $Q_X = Q_Y$, and Q_X, Q_Y, P all lie on l , it suffices

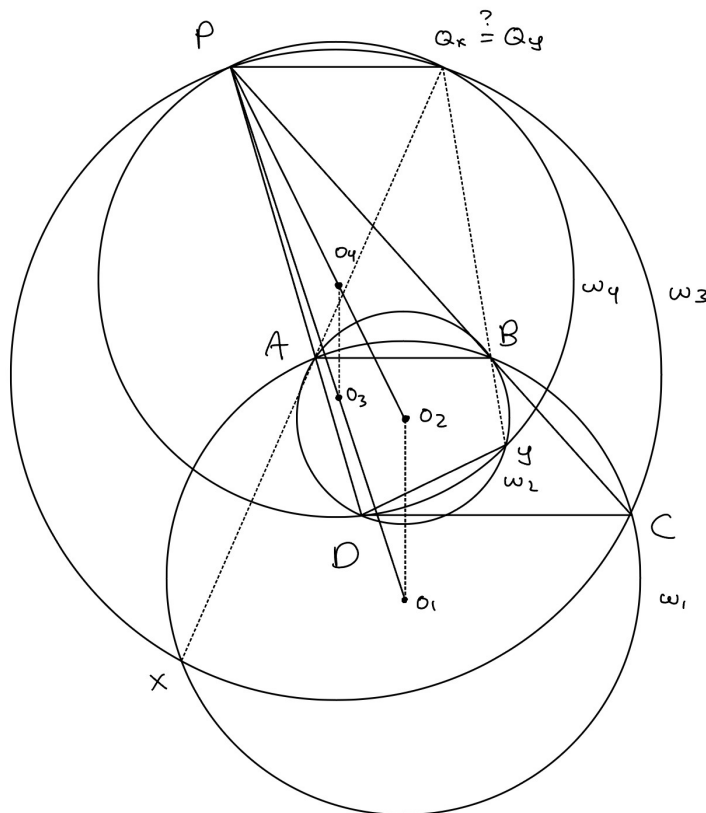
to show that ω_3, ω_4 intersect twice on l , or equivalently that $O_3O_4 \perp AB$. Since O_1O_3 and O_2O_4 both pass through P , it suffices to prove that

$$\frac{O_3P}{O_3O_1} = \frac{O_4P}{O_4O_2}.$$

But we have

$$\frac{O_3P}{O_3O_1} = \frac{O_3X}{O_3O_1} = \frac{PX}{PB} = \frac{PC}{PB} = \frac{PA}{PD} = \frac{PA}{PY} = \frac{O_4Y}{O_4O_2} = \frac{O_4P}{O_4O_2},$$

where the second and second-last equalities are by Salmon's theorem. Hence we are done.



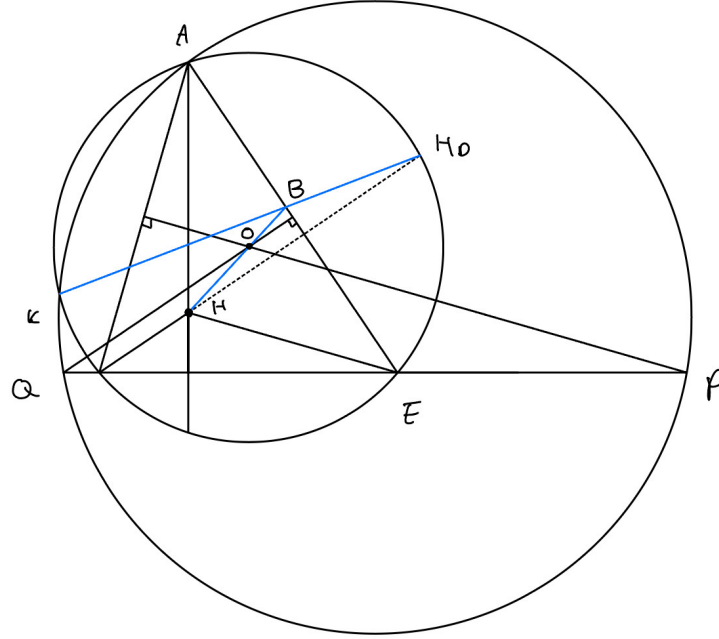
(MOP HW 2025 G5) Let ADE be a triangle with circumcenter O . P and Q are on DE such that $AP = PD$ and $AQ = QE$. Let $(APQ) \cap (ADE) = \{A, K\}$. Moreover, consider $B \in AE$, $C \in AD$ such that

$$\angle AKB = \angle AKC = 90^\circ - \angle DAE.$$

Prove that B, O, C collinear.

Solution 6.4. Here's the solution without $\sqrt{bc}$ inversion. Let H be the orthocenter of (ADE) , and let $KB \cap (ADE) = K_D, KC \cap (ADE) = K_E$. Then $\angle H_D K A = 90^\circ - \angle DAE = \angle HDA$, so H_D is the reflection of H across AE . Similarly H_E is the reflection of H across AD . We claim K lies on the reflections of OH across AE and AD ; this would imply the problem, as then B and C both lie on OH and their respective reflections.

To this end, let X and Y be the feet of the altitudes from K to AD and AE , respectively. Let l be the dilation of XY about K by a factor of 2. Then by well-known Simson line properties H lies on l , and we wish to prove that so does O . Hence it suffices to prove that XY and OH are parallel.



For this, let (ODE) intersect AE and AD at P' and Q' , respectively. By elementary angle chasing we find P' is on OP and Q' on OQ . Now by inversion about O we find that $AP'Q'K$ is cyclic. Hence by spiral similarities, $KDQ' \sim KEP'$. Now, we prove the following lemma.

Lemma 6.5. *Consider two arbitrary points U, V in the same plane as A, D, E . Suppose the perpendiculars from U, V to AE intersect AD at U_1, V_1 (respectively), while the perpendiculars from U, V to AD intersect AE at U_2, V_2 (respectively). Prove that the orientation of UV depends only on the (signed) ratio of $|U_1V_1|$ to $|U_2V_2|$.*

Proof. Coordinates work. Alternatively, for a more “synthetic” solution, let the projections of UV to AD and AE have lengths x and y , respectively. Then the given ratio is a fixed multiple of the ratio between x and y . Now let v be a unit vector in the direction of AD , w a unit vector in the direction of AE , and let r denote the vector $\vec{UV}$. Then $x = v \cdot r$ and $y = w \cdot r$, so the ratio of x to y clearly determines the orientation of r by elementary linear algebra. $\square$

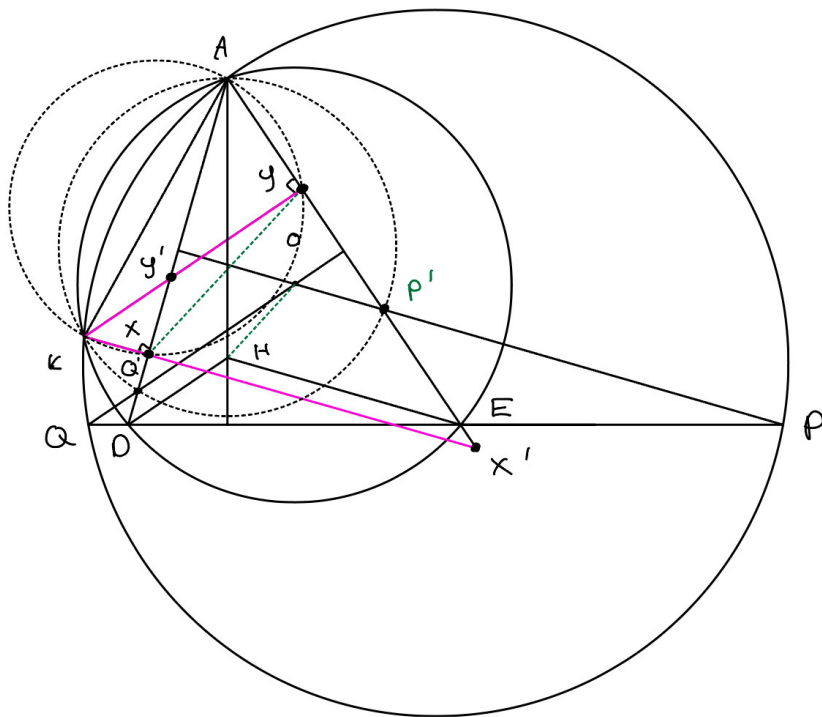
With this lemma in hand, let $X' = KX \cap AE$ and $Y' = KY \cap AD$. Then we wish to show

$$\frac{XY'}{YX'} = \frac{DQ'}{P'E}.$$

But we have

$$\frac{XY'}{YX'} = \frac{KX}{KY} = \frac{KD}{KE} = \frac{DQ'}{EP'}$$

where the last equality is by the spiral similarity $KDQ' \sim KEP'$. Hence we are done.



Here's the solution with $\sqrt{bc}$ inversion about A in ADE . I'll let you check on your own that this is the resulting problem.

Consider a triangle APQ with incenter O . Let $D = OQ \cap (APQ)$, $E = OP \cap (APQ)$, and $K = DE \cap PQ$. Let the perpendicular from K to AE intersect AD at B , and let the perpendicular from K to AD intersect AE at C . Prove $(ABCO)$ is cyclic.

For this solution, ignore the configuration issue disclaimer from earlier. Let $F = AO \cap (ADE)$. Also consider $Y, B' \in AB$ and $Z, C' \in AC$ such that $KA = KY = KZ$, $KB = KB'$, and $KC = KC'$. We first remove the $(ABCO)$ condition: observe by definition that $AOYZ$ is cyclic, so by spiral similarities it suffices to show

$$\frac{BY}{CZ} = \frac{OY}{OZ}.$$

Observe that $\angle OZY = \angle OAY = \angle FED$, and similarly $\angle OYZ = \angle FDE$. So it suffices to show

$$\frac{BY}{CZ} = \frac{FD}{FE},$$

or alternatively

$$\frac{AB'}{AC'} = \frac{FD}{FE}.$$

Now let $DP \cap EQ = T$. Observe that

$$\angle TDF = 180^\circ - \angle FDP = 180^\circ - \angle FEP = \angle DFE + 90^\circ = \angle BAC + 90^\circ = 180 - \angle KBA = \angle KB'A,$$

and similarly $\angle TEF = \angle KC'A$. We claim $KB'AC'$ and $TDFE$ are directly similar; this would solve the problem. We have already checked all four of their angles are equal, so it suffices to show that $\angle KAD =$

$\angle TFD$, or alternatively that KA and TF intersect on (FED) . But letting $X = TF \cap (FED)$ and $X' = KA \cap (FED)$, we have that

$$[X, E; Q, D] =^T [F, Q; E, P] =^O [A, D; P, E] =^K [X', E; Q, D],$$

so $X = X'$, as desired.

